

# CP020003 — Artificial Intelligence 2026

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## Week 12— Homework Assignment

In class this week (Week 12) we crossed over from discriminative AI ("what is in this image?") into generative AI ("make something that didn't exist a second ago"). We watched an LLM generate text token-by-token and a diffusion model denoise a whole image at once, then worked through text-to-image generation, hyperparameter tuning (steps and guidance scale) with CLIP Score, image captioning, an image→text→image round trip, inpainting, and 1–4 step "turbo" distillation — all using Stable Diffusion v1.5, SD-Turbo, Qwen2.5-0.5B, BLIP, and CLIP.

This assignment asks you to go explore the wider, fast-moving world of generative AI on your own: pick ONE modern generative task, then find and try FIVE different modern models for that task, and compare what you get.

### 1. Choose Your Task 🎯

Pick exactly ONE generative AI task. You are not limited to the tasks covered in class (text-to-image, image-to-text, inpainting) — this week's goal is to explore new, modern tasks and models. Some options, with example current models to get you started:

| Task Category                          | Example Modern Models (2024–2026)                                                                         | Typical Input → Output        |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------|
| Text → Image                           | Stable Diffusion 3 / SDXL, FLUX.1 (dev/schnell), Playground v2.5, DALL·E 3, Ideogram 2.0, Google Imagen 3 | prompt → image                |
| Text → Video                           | OpenAI Sora, Runway Gen-3, Kling, Luma Dream Machine, Pika, CogVideoX, Hunyuan Video                      | prompt → short video clip     |
| Image → Video                          | Stable Video Diffusion, I2VGen-XL, Kling Image-to-Video, Runway Gen-3 (image mode)                        | still image → animated clip   |
| Text → Speech (TTS)                    | ElevenLabs, XTTS-v2, Bark, F5-TTS, OpenVoice, MeloTTS                                                     | text → spoken audio           |
| Speech → Text (ASR)                    | Whisper (large-v3 / turbo), faster-whisper, wav2vec2, NVIDIA Canary, Distil-Whisper                       | audio → transcript            |
| Text → Music / Audio                   | Suno, Udio, MusicGen, Stable Audio Open, AudioCraft                                                       | prompt → music/sound clip     |
| Text → 3D                              | Meshy, Tripo3D, Shap-E, TripoSR, Stable Fast3D                                                            | prompt or image → 3D model    |
| Image → Text (Captioning / VQA)        | BLIP-2, LLaVA-1.6, Qwen2-VL, InternVL2, MiniCPM-V, GPT-4V                                                 | image (+question) → text      |
| Image → Image (Edit / Inpaint / Style) | InstructPix2Pix, FLUX Kontext, IP-Adapter, ControlNet, SDXL Inpainting                                    | image + prompt → edited image |
| Text → Text (LLM Chat/Reasoning)       | Llama 3.1/3.2, Qwen2.5, Gemma 2, Mistral Small/Large, Phi-3.5, DeepSeek-V2                                | prompt → generated text       |

*Not sure where to start? Pick whichever row genuinely interests you (a product you use, a hobby, a research question), or ask your instructor. You are also free to propose a task that isn't in this table if it's clearly a generative AI task.*

State which task you picked and, in 1–2 sentences, why it interests you.

## 2. Find and Run 5 Modern Models 🔍

For your chosen task, find FIVE different modern generative models and run all of them on the same input (same prompt, same image, same audio clip, etc.) so the comparison is fair. Good places to find and try models:

- Hugging Face Hub (models) and Hugging Face Spaces (free hosted demos — great when a model is too big for Colab's free GPU)
- Hugging Face diffusers / transformers libraries (same pattern we used in class — swap the model ID)
- Official model playgrounds or free tiers (e.g. a provider's web demo or API free credits)
- Papers with Code / model leaderboards for your task, to see what's considered "modern" right now

It's completely fine to mix: run some models locally in Colab (open-source, fits a free T4), and try others through a free web demo or API if they're too large or closed-source to run yourself. Just be clear in your report about how you ran each one.

**Report, for each of the 5 models:**

- Model name, and where you ran it (Colab / HF Space / API / other)
- Open-source or closed-source, and approximate size if known (e.g. parameter count)
- The exact input you gave it (prompt / image / audio)
- The output it produced (embed the image, paste the text, link/describe the audio-video, etc.)
- How long it took to generate (wall-clock time), where measurable

## 3. Build & Evaluate Your Comparison 🔧

Follow the same spirit as the in-class notebook:

- Setup — briefly describe each of the 5 models and how you accessed it.
- Run — generate outputs from all 5 models on the same input(s); show them side by side.
- Score — use at least one real metric appropriate to your task (see table below) in addition to your own qualitative judgment. Reuse CLIP Score from class where it applies (text-to-image, image-to-text, image editing).
- Compare — which model produced the best result, which was fastest, which was most reliable? Include at least one case where a model failed or produced a poor result.
- Conclusion — what surprised you, and which model would you actually choose for a real use case, and why.

| Task Category                | Common Headline Metric(s)                                                         | Notes                                                     |
|------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------|
| Text → Image                 | CLIP Score, human preference / A-B vote                                           | Same metric as this week's Section 2–3                    |
| Text → Video / Image → Video | CLIP Score per frame, motion smoothness (qualitative), frame consistency          | Video metrics are harder — qualitative comparison is fine |
| Text → Speech                | Mean Opinion Score (MOS, human 1–5 rating), Word Error Rate of a re-transcription | Re-transcribe with Whisper and compare to original text   |
| Speech → Text                | Word Error Rate (WER), Character Error Rate (CER)                                 | Use a dataset with ground-truth transcripts               |
| Text → Music/Audio           | Human preference, prompt-adherence rating                                         | No single standard automatic metric yet                   |
| Text → 3D                    | Visual inspection from multiple angles, human rating                              | Automatic 3D metrics are out of scope for this course     |

| Task Category                                                                                                                                                                                                                                                                                                                                                                                                                                              | Common Headline Metric(s)                                                             | Notes                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Image → Text (Captioning/VQA)                                                                                                                                                                                                                                                                                                                                                                                                                              | CLIP text–image similarity, BLEU/CIDEr (if ground-truth captions exist)               | Same idea as this week's Section 3                          |
| Image → Image (Edit/Inpaint)                                                                                                                                                                                                                                                                                                                                                                                                                               | CLIP score inside edited region, pixel MSE outside region                             | Same as this week's Section 5                               |
| Text → Text (LLM)                                                                                                                                                                                                                                                                                                                                                                                                                                          | Perplexity (if accessible), human judgment on correctness/coherence, response latency | Perplexity needs access to logits — open-source models only |
| <p><b>⚠ Grading Emphasis — Read This Carefully</b></p> <p>Getting five demos to simply run is not enough. Most marks come from: choosing a real, current generative AI task and genuinely different modern models (not five checkpoints of the same base model), reporting a proper metric (not just "it looks nice"), showing actual outputs for every model (including at least one failure case), and writing a clear, comparison-based conclusion.</p> |                                                                                       |                                                             |

#### 4. Bonus Points ✨

Apply a "fast / distilled" version of your task if one exists (in the spirit of Section 6 this week — SD-Turbo, LCM, or an equivalent turbo/distilled model for your chosen task) and compare its speed and quality against one of your five "full" models — +2% on your raw grade (out of 100%) for grading.

*This bonus applies on top of Section 3 above — you still need to properly run and evaluate your five main models.*

#### Submission

Submit your notebook (.ipynb) with all cells run and outputs visible (or a short PDF/slide export if you used web demos/APIs for some models), plus a short written summary (task chosen, the 5 models, metric(s) used, and your conclusion) of no more than one page.